THEO GUEGAN

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Experience

Founding Engineer, Embodied AI **Feb 2026 – Present**
Opalin (Claire Delaunay, ex-VP NVIDIA Robotics) *Redwood City, CA*

- Teleop-to-fleet pipeline: data from multiple teleoperators gathered, filtered and evaluated under the supervision of a single operator, within a day; down from days and weeks per experiment.
- VLAs and world action models: architectures, training, fine-tuning and evaluation of visuomotor policies over multi-view video and proprioception, run in real time on the robots.
- Data curation: 3D dataset viewer; per-task statistics (3D workspace and grasp-point coverage, phase durations, smoothness) and robust outlier detection (MAD, MCD, DTW novelty, near-duplicates); half the data, success 50% to 98%.
- Real-time arm control with Pinocchio IK/FK and impedance control to 0.1 mm over CAN.

Embedded AI Software Engineer **Sep 2024 – Feb 2025**
Thales Land & Air Systems *Vélizy-Villacoublay, France*

- Fully decentralized embedded AI for drone swarm coordination and control, removing the need for a permanent datalink to the ground station.
- Embedded Lua script engine and on-device LLM voice-to-script generation (98% correctness) for rapid mission prototyping (Rust harness, end-to-end deployment).
- Cut new-mission deployment time by 10x and improved reliability and predictability by 4x.
- Part of Thales' COHESION demonstrator (JDEC 2024), featured in French media.

Research Assistant, Autonomous Vehicle Control Lead **Feb 2024 – Jun 2025**
CNRS Hauts-de-France, Heudiasyc Lab *Compiègne, France*

- Control architecture for a fleet of shared autonomous vehicles in dense urban environments; replaced rigid trajectory planning with waypoint-based navigation.
- Led sim-to-real deployment of the control stack onto a Renault Zoe (ROS/Python): waypoint control law, adaptive cruise control, obstacle avoidance; 99% safety in simulation.
- 1st School Award (2024), 1st place Open Category (2025), UTAC Challenge.

Research

Behavior Cloning of MPC for 3-DOF Robotic Manipulators | arXiv:2606.00383

- Poster at the Workshop on RL in the Era of Imitation Learning, IEEE ICRA 2026.
- Distilled an MPC policy into neural surrogates: 3x lower inference latency at 85% success; static architectures outperform temporal ones.
- Ablation and statistical studies over traditional ML algorithms.

Skills

Learning: imitation learning, LeRobot, VLAs, JEPA and latent-predictive objectives, multimodal video architectures, flow matching, LoRA/PEFT
Robot systems: multi-input teleoperation devices, data curation pipelines, real-time control, fleet deployment, Pinocchio IK/FK, RNEA, impedance control, MPC, ROS2, dora-rs, RTOS, CAN
Scale: multi-GPU training, efficient inference, PyTorch, Docker, CI/CD, Dagster, DuckDB, Lance
Languages & sim: Python 3.12+, Rust, C++17, Zig, Lua, MATLAB, Bash; mujoco, genesis

Education

University of Waterloo **Waterloo, Canada**
M.Sc. Artificial Intelligence, AI/ML Sep 2025 – Dec 2025

- GPA 4.0/4.0: Deep Learning (Bryan Tripp), Machine Learning (Terrence C. Stewart), Image Processing.

Université de Technologie de Compiègne **Compiègne, France**
M.Sc. Computer Engineering, Embedded & Autonomous Systems Sep 2021 – Sep 2026

- GPA 5.0/5.0: Robotics Control, Embedded Systems, Kinematics.

Hackathons

- 2nd prize, Seeed x NVIDIA x Hugging Face Embodied AI Hackathon (2026); 2nd prize, Open World Hackathon by VLGE (2026); Winner, UTC x ST2I Hackathon.

Open Source

- fathom: zero-overhead Rust visualization of multimodal spatiotemporal data (video, depth, point clouds, trajectories).
- drover: Rust agent runtime running coding agents in git worktrees, driven from the terminal or Discord.
- Reachy Mini and SO-ARM-101 manipulation stacks; peillute, distributed P2P payments in Rust; Kaggle Brain-to-Text '25.